e itself: edges parallel to e become loops at v_e, while edges xv and yv become parallel edges between v_e and v (Fig. 1.10.1). Thus, formally, $E(G/e) = E \setminus \{e\}$, and only the incidence map $e' \mapsto \{\text{init}(e'), \text{ter}(e')\}$ of G has to be adjusted to the new vertex set in G/e. Contracting a loop thus has the same effect as deleting it.

The notion of a minor adapts accordingly. The contraction minor G/P defined by a partition P of $V(G)$ into connected sets has precisely those edges of G that join distinct partition classes. If there are several such edges between the same two classes, they become parallel edges of G/P. However, we do not normally give G/P any loops resulting from edges of G whose ends lie in the same partition class U. This would require us to say which of the edges of $G[U]$ are contracted (assuming they induce a connected spanning subgraph of $G[U]$), or at least how many are, which seems futile if we do not care about loops in G/P anyway.

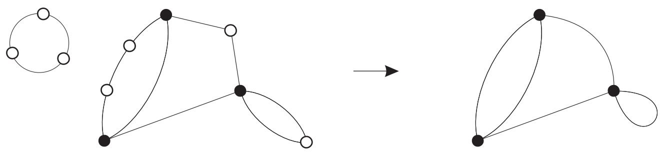

Fig. 1.10.2. Suppressing the white vertices

suppressing a vertex

If v is a vertex of degree 2 in a multigraph G, then by *suppressing* v we mean deleting v and adding an edge between its two neighbours. (If its two incident edges are identical, i.e. form a loop at v, we add no edge and obtain just $G - v$. If they go to the same vertex $w \neq v$, the added edge will be a loop at w. See Figure 1.10.2.) Since the degrees of all vertices other than v remain unchanged when v is suppressed, suppressing several vertices of G always yields a well-defined multigraph that is independent of the order in which those vertices are suppressed.

Finally, it should be pointed out that authors who usually work with multigraphs tend to call them 'graphs'; in their terminology, our graphs would be called 'simple graphs'.

Exercises

1.$^-$ What is the number of edges in a K^n?

2. Let $d \in \mathbb{N}$ and $V := \{0,1\}^d$; thus, V is the set of all 0–1 sequences of length d. The graph on V in which two such sequences form an edge if and only if they differ in exactly one position is called the *d-dimensional cube*. Determine the average degree, number of edges, diameter, girth and circumference of this graph.

   (Hint for the circumference: induction on d.)

3. Let G be a graph containing a cycle C, and assume that G contains a path of length at least k between two vertices of C. Show that G contains a cycle of length at least $\sqrt{k}$.

4.$^-$ Is the bound in Proposition 1.3.2 best possible?

5. Let v_0 be a vertex in a graph G, and $D_0 := \{v_0\}$. For $n = 1, 2, \ldots$ inductively define $D_n := N_G(D_0 \cup \ldots \cup D_{n-1})$. Show that $D_n = \{ v \mid d(v_0, v) = n \}$ and $D_{n+1} \subseteq N(D_n) \subseteq D_{n-1} \cup D_{n+1}$ for all $n \in \mathbb{N}$.

6. Show that $\mathrm{rad}(G) \leqslant \mathrm{diam}(G) \leqslant 2\,\mathrm{rad}(G)$ for every graph G.

7. Prove the weakening of Theorem 1.3.4 obtained by replacing average with minimum degree. Deduce that $|G| \geqslant n_0(d/2, g)$ for every graph G as given in the theorem.

8. Show that graphs of girth at least 5 and order n have a minimum degree of $o(n)$. In other words, show that there is a function $f\colon \mathbb{N} \to \mathbb{N}$ such that $f(n)/n \to 0$ as $n \to \infty$ and $\delta(G) \leqslant f(n)$ for all such graphs G.

9.$^+$ Show that every connected graph G of order at least 3 contains a path or cycle of length at least $\min\{2\delta(G), |G|\}$.

10. Show that a connected graph of diameter k and minimum degree d has at least about $kd/3$ vertices but need not have substantially more.

11.$^-$ Show that the components of a graph partition its vertex set. (In other words, show that every vertex belongs to exactly one component.)

12.$^-$ Show that every 2-connected graph contains a cycle.

13. Determine $\kappa(G)$ and $\lambda(G)$ for $G = P^m,\ K^m,\ K_{m,n},\ C^n$ and the d-dimensional cube (Exercise 2), for all $m \geqslant 1$ and $d, n \geqslant 3$.

14.$^-$ Is there a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, for all $k \in \mathbb{N}$, every graph of minimum degree at least $f(k)$ is k-connected?

15. Let α, β be two graph invariants with positive integer values. Formalize the two statements below, and show that each implies the other:

 (i) β is bounded above by a function of α;

 (ii) α can be forced up by making β large enough.

 Show that the statement

 (iii) α is bounded below by a function of β

 is not equivalent to (i) and (ii). Which small change will make it so?

16. Show that every graph that is k-edge-connected but loses this property whenever we delete an edge has a vertex of degree k.

17.$^+$ Show for every $k \in \mathbb{N}$ that every graph of minimum degree $2k$ has a $(k+1)$-edge-connected subgraph. Is it enough to assume an average degree of at least $2k$?

18. Consider the proof of Theorem 1.4.3. Would it not seem more natural to assume in the second statement of ($*$) that $\varepsilon(G') > \gamma - k$, as required for H in the statement of the theorem?

    (i) Look how this alteration would change the proof: which parts would carry over, which could be adapted, and which would fail?

    (ii) Explain how the use of an assumption of the form $m \geqslant c_k n - b_k$ rather than $m \geqslant c_k n$ helps to obtain a contradiction in the final inequality of the proof.

19. Prove Theorem 1.5.1.

20. Revisit the proof that every connected graph has a spanning tree given as an application of Theorem 1.5.1 (iii). What is wrong with the analogous 'proof' that says, 'take a maximal acyclic subgraph and apply (iv)'?

    (Hint: Something must be wrong, as the 'proof' does not use the assumption that the graph is connected. But where exactly is the error?)

21.$^-$ (i) Show that every tree T has at least $\Delta(T)$ leaves.

    (ii) Deduce that in a connected graph G we can delete $\Delta(G)$ vertices so that the rest remains connected.

22. Let T be a tree with $\ell \geqslant 2$ leaves and maximum degree at most 3.

    (i)$^-$ Show that T has exactly $\ell - 2$ vertices of degree 3.

    (ii) Show that T contains $\lfloor \ell/2 \rfloor$ disjoint paths between distinct leaves.

23. Find two very short proofs, one by induction and another without, that every tree has more leaves than vertices of degree at least 3.

24. Let F, F' be forests on the same set of vertices, with $\|F\| < \|F'\|$. Show that F' has an edge $e \notin F$ such that $F + e$ is again a forest.

25. Show that the tree-order associated with a rooted tree T is indeed a partial order on $V(T)$, and verify the claims made about this partial order in the text.

26. Show that a graph is 2-edge-connected if and only if it has a *strongly connected* orientation, one in which every vertex can be reached from every other vertex by a directed path.

27. Modify the proof of Proposition 1.5.5 to show that we may specify any vertex of G as the root of the normal spanning tree sought.

28. (i) Find a short proof for the existence of normal spanning trees in connected graphs that applies induction by deleting a vertex.

    (ii) Adapt your proof to show that every path in a connected graph extends to a normal spanning tree.

29.$^+$ Let G be a connected graph, and let $r \in G$ be a vertex. Starting from r, move along the edges of G, going whenever possible to a vertex not visited so far. If there is no such vertex, go back along the edge by which the current vertex was first reached (unless the current vertex is r; then stop). Show that the edges traversed form a normal spanning tree in G with root r.

(This procedure has earned those trees the name of *depth-first search* trees.)

30. Let $\mathcal{T}$ be a set of subtrees of a tree T, and $k \in \mathbb{N}$.

(i) Show that if the trees in $\mathcal{T}$ have pairwise non-empty intersection then their overall intersection $\bigcap \mathcal{T}$ is non-empty.

(ii) Show that either $\mathcal{T}$ contains k disjoint trees or there is a set of at most $k-1$ vertices of T meeting every tree in $\mathcal{T}$.

31. Show that every automorphism of a tree fixes a vertex or an edge.

32.$^-$ Do the partition classes of a regular bipartite graph always have the same size?

33.$^-$ Show that a graph is bipartite if and only if every *induced* cycle has even length.

34. Is the vertex partition of a bipartite graph uniquely determined?

35. Prove or disprove that a graph is bipartite if and only if no two adjacent vertices have the same distance from any other vertex.

36. Proposition 1.6.1 characterizes the graphs that contain no odd cycle. Can you characterize those that contain no even cycle?

37. Find a function $f: \mathbb{N} \to \mathbb{N}$ such that, for all $k \in \mathbb{N}$, every graph of average degree at least $f(k)$ has a bipartite subgraph of minimum degree at least k.

38. Show that the minor relation $\preccurlyeq$ defines a partial ordering on any set of pairwise non-isomorphic finite graphs. Is the same true for infinite graphs?

39. If we had been careless, we might have defined a walk as an alternating sequence of vertices and edges, $v_0 e_0 v_1 e_1 \ldots e_{k-1} v_k$ say, such that every edge e_i is incident with both v_i and v_{i+1}. Show that Theorem 1.8.1 would fail with this definition, and find where the difference between the two definitions matters in the proof.

40. Prove or disprove that every connected graph contains a walk that traverses each of its edges exactly once in each direction.

41. Show that a cut in a connected graph G is a bond if and only if both parts of the corresponding bipartition of $V(G)$ are connected in G. Is every atomic cut a bond?

42. Let A be a set of vertices in a tree T. Show that T contains a set of edge-disjoint paths with ends in A such that every vertex from A except at most one is the end of exactly one such path.

43. To how few edges can you reduce the complete graph K^n by deleting one edge at a time from a 4-cycle found in the current graph?

44. Show that the cycle space of a graph is spanned by

    (i) its induced cycles;

    (ii) its geodesic cycles.

    (A cycle $C \subseteq G$ is *geodesic* in G if, for every two vertices of C, their distances in G equals their distance in C.)

45. Show that the set of sides of a cut in a connected graph is well defined.

46.$^-$ Show directly, without appealing to atomic cuts, that the cuts of a graph together with the empty set form a subspace of its edge space. How does the vertex partition of a sum of two given cuts arise from their vertex partitions?

47. Let F be a cut in G, with vertex partition $\{V_1, V_2\}$. For $i = 1, 2$ let $C^i_1, \dots, C^i_{k(i)}$ denote the components of $G[V_i]$. Use the C^i_j to define bonds whose disjoint union is F.

48. Show that the cut space of any graph has a basis of atomic cuts.

49. Prove that the cycles and the cuts in a graph together generate its entire edge space, or find a counterexample.

50.$^-$ Show the following duality between the fundamental cycles C_e and the fundamental cuts D_f in a graph with respect to some fixed spanning tree: $e \in D_f \Leftrightarrow f \in C_e$.

51. Show that in a connected graph the minimal edge sets containing an edge from every spanning tree are precisely its bonds.

52. Given a spanning tree $T = (V, F)$ of a connected graph $G = (V, E)$, its fundamental cuts D_f and fundamental cycles C_e witness the following:

    (i) $\mathcal{B}$ has a basis $(\, D_f \mid f \in F \,)$ such that $D_f \cap F = \{f\}$ for all $f \in F$;

    (ii) $\mathcal{C}$ has a basis $(\, C_e \mid e \in E \smallsetminus F \,)$ such that $C_e \cap (E \smallsetminus F) = \{e\}$ for all $e \in E \smallsetminus F$.

    Show that, conversely, every set $F \subseteq E$ satisfying (i) and (ii) is the edge set of a spanning tree of G. Is the same true as soon as F satisfies one of these two statements? Then show that this spanning tree has precisely the D_f as fundamental cuts, and the C_e as fundamental cycles.

53. Let F be a set of edges in a graph G.

    (i) Show that F extends to an element of $\mathcal{B}(G)$ if and only if it contains no odd cycle.

    (ii)$^+$ Show that F extends to an element of $\mathcal{C}(G)$ if and only if it contains no odd cut.

54.$^+$ In a graph G let a, b be two vertices that are separated by a cut F of k edges but cannot be separated by fewer edges. Show that F is not a sum of cuts of fewer than k edges.

55.$^+$ Prove that the edge set of any graph G can be written as a disjoint union $E(G) = C \cup D$ with $C \in \mathcal{C}(G)$ and $D \in \mathcal{B}(G)$.

56. Show that a set of vertices lies in the image of the incidence matrix of a connected graph if and only if it has even cardinality.

57. (i) Generalize Proposition 1.9.6 by describing the images under B and $B^\top$ of given sets $F \subseteq E$ and $U \subseteq V$, as indicated in the text.

    (ii) Reprove Proposition 1.9.7 for matrices with values in $\mathbb{F}_2$ by showing that $BB^\top$ and $A+D$ define the same the maps $\mathcal{V} \to \mathcal{V}$.

58. Let $A = (a_{ij})_{n\times n}$ be the adjacency matrix of the graph G. Show that the matrix $A^k = (a'_{ij})_{n\times n}$ displays, for all $i, j \leqslant n$, the number a'_{ij} of walks of length k from v_i to v_j in G.

Notes

The terminology used in this book is mostly standard. Alternatives do exist, and some of these are stated when a concept is first defined.

Our formal definition of a graph $G = (V, E)$ with $E \subseteq [V]^2$ is intended to convey two messages: that the edges are undirected (since $\{u, v\} = \{v, u\}$ for sets), and that there are neither loops (since $\{v, v\} \notin [V]^2$ because $|\{v, v\}| = 1$) nor multiple edges (since two sets are equal as soon as they have the same elements). This formal definition – like any other – occasionally clashes with other standard terminology.[13] But avoiding all such possible clashes would make the terminology so unwieldy that it would defeat the purpose of clarity.

There is one small point where our notation deviates slightly from standard usage. Complete graphs, paths, cycles etc. of given order are usually denoted by K_n, P_k, C_ℓ and so on, but we use superscripts instead of subscripts. This has the advantage of leaving the variables K, P, C etc. free for ad-hoc use: we may now enumerate components as $C_1, C_2, \dots$, speak of paths $P_1, \dots, P_k$, and so on – without any danger of confusion.

Theorem[14] 1.3.4 was proved by N. Alon, S. Hoory and N. Linial, The Moore bound for irregular graphs, *Graphs Comb.* **18** (2002), 53–57. The proof uses an ingenious argument counting random walks along the edges of the graph considered.

The main assertion of Theorem 1.4.3, that an average degree of at least $4k$ forces a k-connected subgraph, is from W. Mader, Existenz n-fach zusammenhängender Teilgraphen in Graphen genügend großer Kantendichte, *Abh. Math. Sem. Univ. Hamburg* **37** (1972) 86–97.

For the history of the Königsberg bridge problem, and Euler's actual part in its solution, see N.L. Biggs, E.K. Lloyd & R.J. Wilson, *Graph Theory 1736–1936*, Oxford University Press 1976.

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[13] For example, when $e = \{u, v\}$ is an edge of G, then $G - e$ and $G - \{u, v\}$ mean two different things: in $G - e$ we deleted the edge e but kept the vertices u and v, whereas in $G - \{u, v\}$ we deleted the vertices u, v and all their incident edges.

[14] In the interest of readability, the end-of-chapter notes in this book give references only for Theorems, and only in cases where these references cannot be found in a monograph or survey cited for that chapter.